# Hospital Operations & Financial Overview

Encounters, costs, payer coverage and procedures | Jan 2011 to Feb 2022

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Data Notes

FILTERS

**Year**

All

**Encounter Class**

All

**Payer**

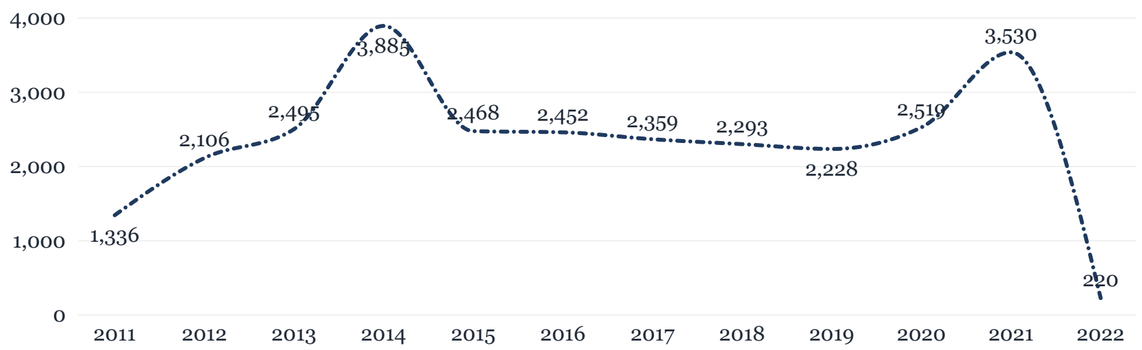
All

Data: Jan 2011 to Feb 2022, one hospital. 2022 is a partial year.

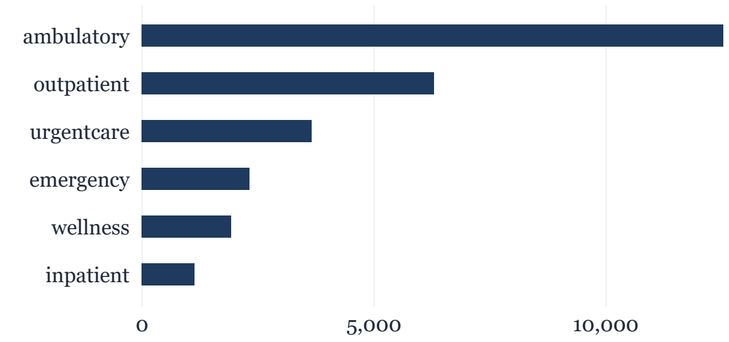
Prepared by Kukoyi Zainab Omotola

| Total Encounters | Total Claim Cost | Zero Coverage % | Payer Coverage % | Avg Claim Cost per Encounter |
|---|---|---|---|---|
| 27,891 | $101.5M | 48.7% | 30.6% | $3,640 |

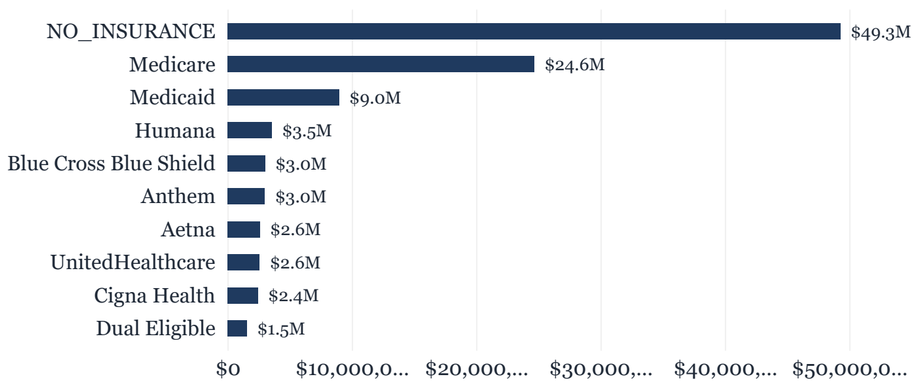
## Encounters by Year



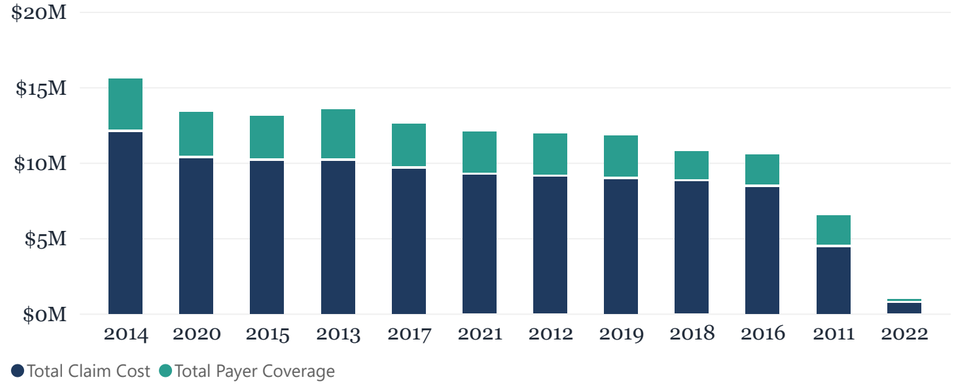

## Encounters by Class




## Claim Cost by Payers




## Claim Cost vs Total Payer Coverage by Year




Footnote;2022 is partial(date ends 5 Feb 2022). Zero payer coverage may mean uninsured, unrecorded or unpaid

# Hospital Operations Encounter Volume Trend

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**Year**

All

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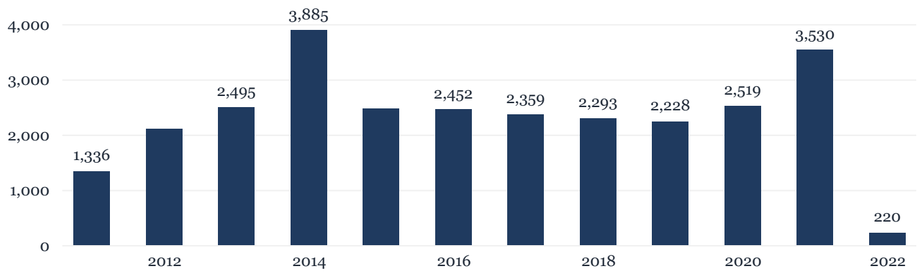
Prepared by Kukoyi Zainab Omotola

Total Encounters
**27,891**

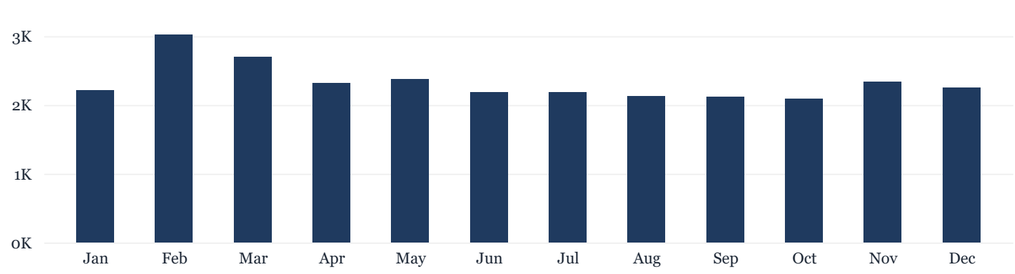
Unique Patients
**974**

Peak month: Feb 2014 (1,179 encounters)

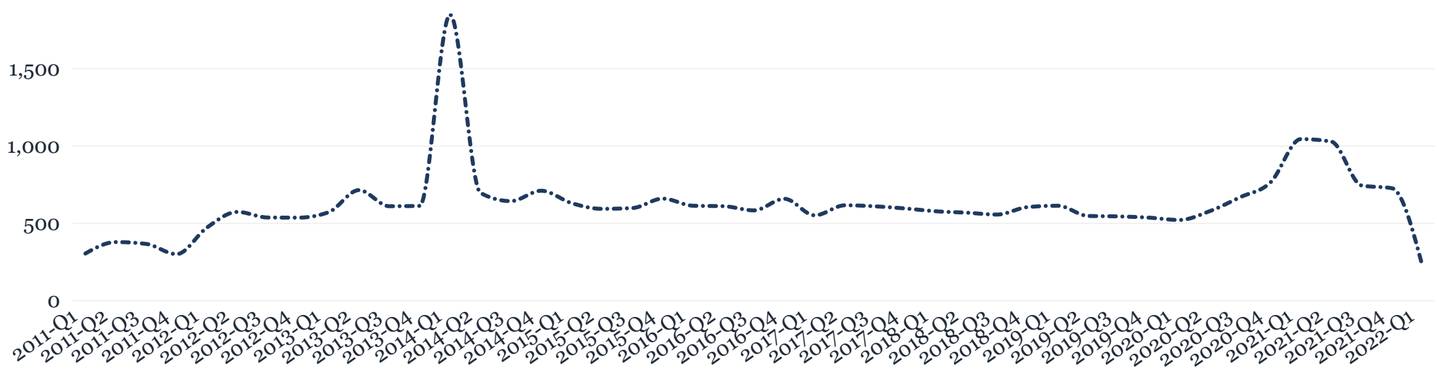
## Encounters by Year




## Encounters by Calendar Month(all years)




## Quarterly Encounters




| Year | Total Encounters | YoY Change % |
|---|---|---|
| 2011 | 1,336 | |
| 2012 | 2,106 | 57.6% |
| 2013 | 2,495 | 18.5% |
| 2014 | 3,885 | 55.7% |
| 2015 | 2,468 | -36.5% |
| 2016 | 2,452 | -0.6% |
| 2017 | 2,359 | -3.8% |
| 2018 | 2,293 | -2.8% |
| 2019 | 2,228 | -2.8% |
| 2020 | 2,519 | 13.1% |
| 2021 | 3,530 | 40.1% |
| 2022 | 220 | |

Footnote;2022 is partial(date ends 5 Feb 2022). February 2014 is an unusually high month (1,179 encounters);the data does not explain why.

# Hospital Operations Encounter Classes Analysis

Encounters, costs, payer coverage and procedures | Jan 2011 to Feb 2022

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FILTERS

**Year**

All

**Encounter Class**

All

**Payer**

All

Data: Jan 2011 to Feb 2022, one hospital. 2022 is a partial year.

Prepared by Kukoyi Zainab Omotola

Total Encounters
**27,891**

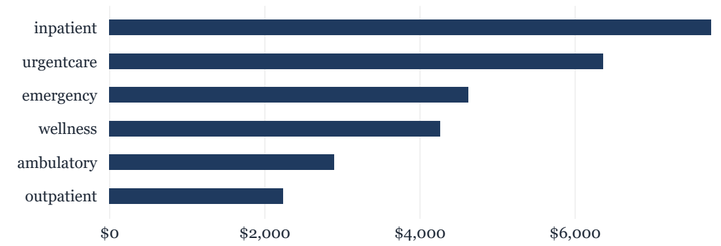
Total Claim Cost
**$101.5M**

Avg Claim Cost per Encounter
**$3,640**

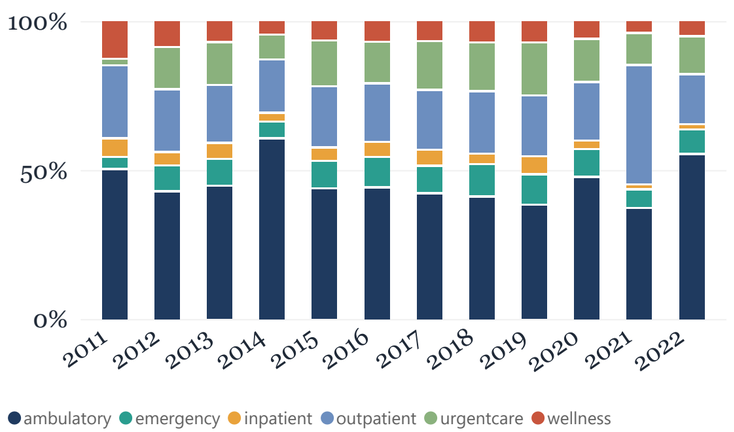
Urgent care is 13.1% of encounters but 23.0% of claim cost. Inpatient has the highest average claim cost ($7,761) and accounts for 58 of the 75 encounters over 24 hours.

| Class | Encounters | Share % | Total Claim Cost | Cost Share % | Avg Cost | Encounters Over 24h | Over 24h % |
|---|---|---|---|---|---|---|---|
| ambulatory | 12,537 | 44.9% | $36.28M | 35.7% | $2,894 | 5 | 0.04% |
| emergency | 2,322 | 8.3% | $10.75M | 10.6% | $4,630 | 5 | 0.22% |
| inpatient | 1,135 | 4.1% | $8.81M | 8.7% | $7,761 | 58 | 5.11% |
| outpatient | 6,300 | 22.6% | $14.09M | 13.9% | $2,237 | 7 | 0.11% |
| urgentcare | 3,666 | 13.1% | $23.35M | 23.0% | $6,369 | 0 | 0.00% |
| wellness | 1,931 | 6.9% | $8.23M | 8.1% | $4,261 | 0 | 0.00% |

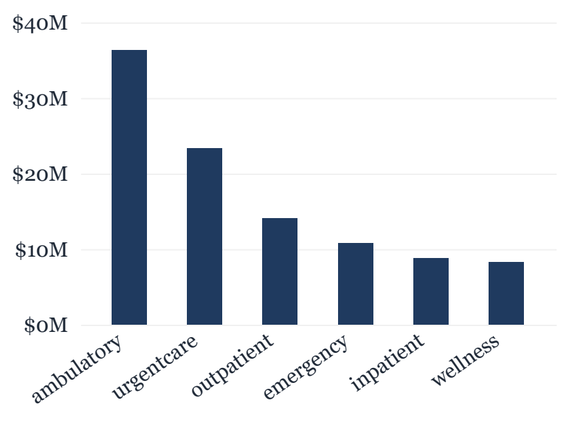
Claim Cost per Encounter



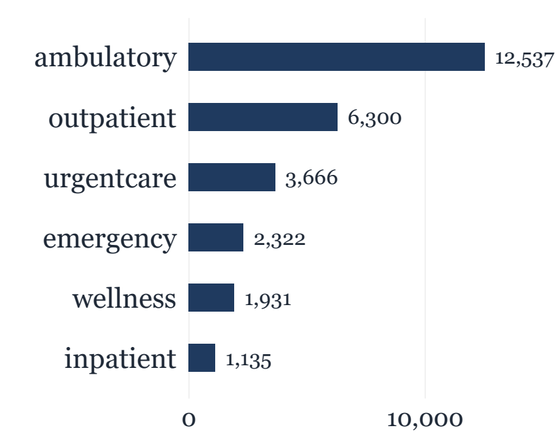

Class Mix by Year




Total Claim Cost by Class




Encounters by Class




Footnote;2022 is partial(data ends 5 Feb 2022). Class definitions are as recorded in the data

# Hospital Operations Cost & Payer Coverage

Encounters, costs, payer coverage and procedures | Jan 2011 to Feb 2022

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- Procedures
- Data Notes

FILTERS

**Year**

All

**Encounter Class**

All

**Payer**

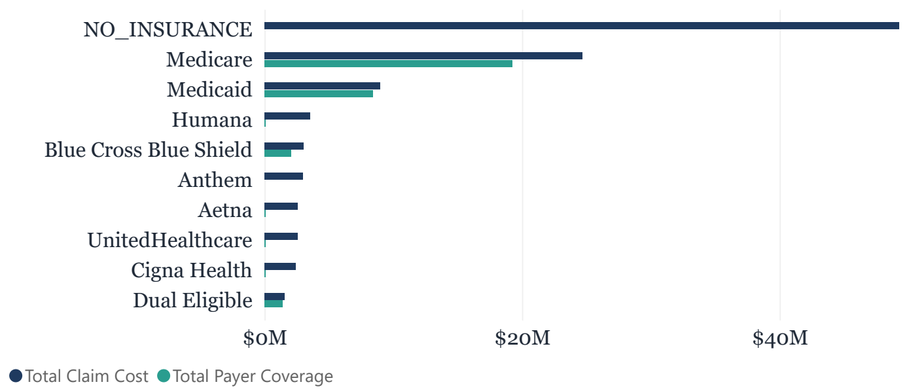
All

Data: Jan 2011 to Feb 2022, one hospital. 2022 is a partial year.

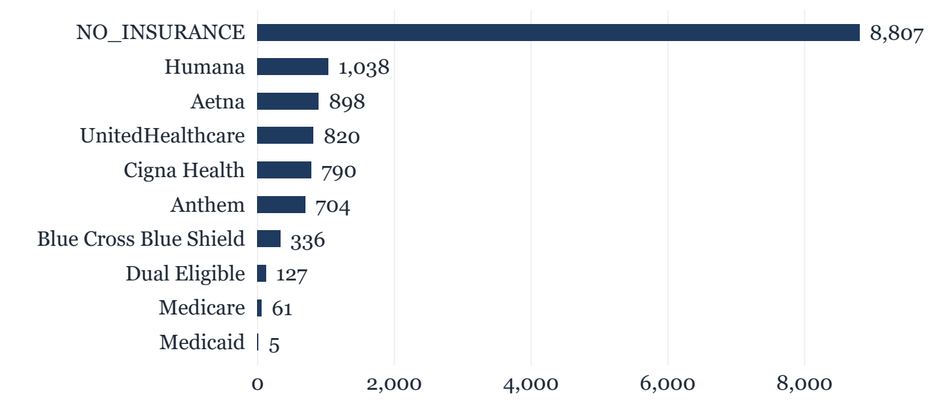
Prepared by Kukoyi Zainab Omotola

| Total Payer Coverage | Payer Coverage % | Zero Coverage % | Cost Not Covered by Payer | Zero Coverage Encounters |
|---|---|---|---|---|
| $31.1M | 30.6% | 48.7% | $70.4M | 13,586 |

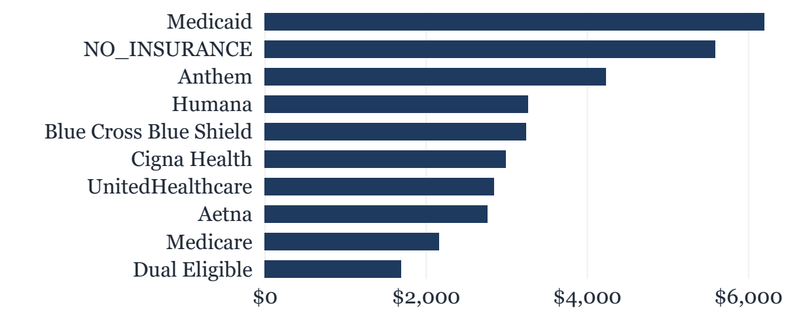
## Claim Cost vs Payer Coverage by Payer



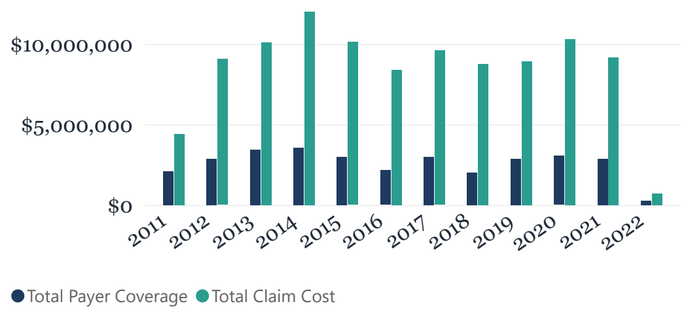

## Zero Coverage Encounters by Payer




## Avg Claim Cost per Encounter by Payer




## Claim Cost vs Payer Coverage by Year




## Key Takeaway

Five private insurers (Humana, Aetna, UnitedHealthcare, Cigna and Anthem) record 0.0% to 0.2% of claim cost as payer coverage, while Medicare, Medicaid and Dual Eligible record 78% to 94%. The data does not explain the difference.

Footnote;Zero payer coverage may reflect uninsured encounters, unrecorded coverage or unpaid claims; the data does not say which. 2022 is partial (data ends 5 Feb 2022).

# Hospital Operations Procedures Utilisation

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FILTERS

**Year**

All

**Encounter Class**

All

**Payer**

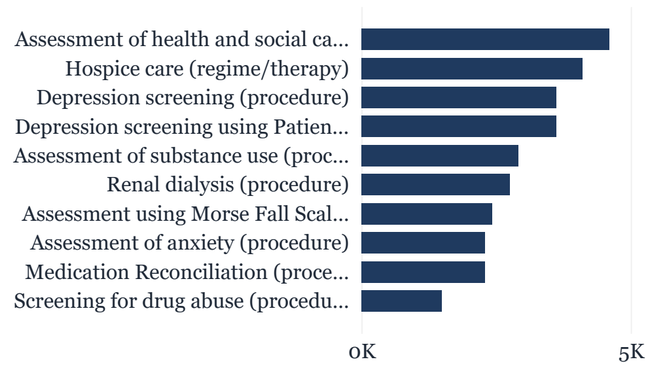
All

Data: Jan 2011 to Feb 2022, one hospital. 2022 is a partial year.

Prepared by Kukoyi Zainab Omotola

| Procedure Count | Avg Procedure Base Cost | Encounters with Procedure | Encounters without Procedure % |
|---|---|---|---|
| 47,701 | $2,212 | 14,670 | 47.4% |

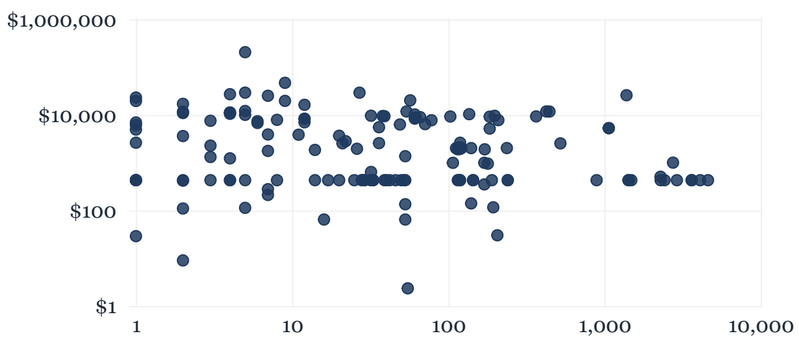
## Top 10 Procedures by Frequency



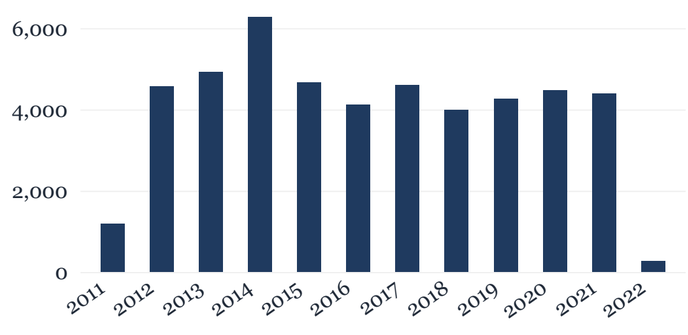

## Highest Average Base Cost (min. 20 records)

| DESCRIPTION | Procedure Count | Avg Procedure Base Cost | Total Procedure Base Cost |
|---|---|---|---|
| Hemodialysis (procedure) | 27 | $29,299.556 | 791,088.00 |
| Electrical cardioversion | 1,383 | $25,903.111 | 35,824,002.00 |
| Percutaneous mechanical thrombectomy of portal vein using fluoroscopic guidance | 57 | $20,228.035 | 1,152,998.00 |
| Colonoscopy | 424 | $11,841.988 | 5,021,003.00 |
| Subcutaneous immunotherapy | 54 | $11,835.500 | 639,117.00 |
| Combined chemotherapy and radiation therapy (procedure) | 444 | $11,796.387 | 5,237,596.00 |
| Cognitive and behavioral therapy (regime/therapy) | 136 | $10,388.081 | 1,412,779.00 |
| Rectal polypectomy | 61 | $10,231.557 | 624,125.00 |
| Induced termination of pregnancy | 32 | $9,641.000 | 308,512.00 |
| Bone density scan (procedure) | 198 | $9,602.864 | 1,901,367.00 |

## Frequency vs Average Base Cost




## Procedures by Year




## Key Takeaway

The ten most frequent procedures make up 63% of records, mostly routine assessments with a flat $431 base cost. Electrical cardioversion has the largest total base cost (1,383 records, average $25,903, about $35.8M). Base cost is not claim cost.

Footnote;Procedure base cost is not the same as claim cost. 47.4% of encounters have no procedure record. Averages use procedures with at least 20 records. 2022 is partial (data ends 5 Feb 2022).

# Hospital Operations Data Notes & KPI Definitions

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All

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Prepared by Kukoyi Zainab Omotola


| KPI name | Definition | Calculation |
|---|---|---|
| Avg Claim Cost per Encounter | Total claim cost divided by number of encounters. | Total Claim Cost / Total Encounters. |
| Avg Procedure Base Cost | Average of procedure base cost. | Sum of BASE_COST / number of procedures. |
| Over 24h % | Encounters where STOP minus START is more than 24 hours, divided by all encounters. | Encounters Over 24h / Total Encounters. Duration = (STOP - START) in hours. |
| Payer Coverage % | Total payer coverage divided by total claim cost. | Total Payer Coverage / Total Claim Cost. |
| Procedure Count | Number of procedure records. | Count of rows in procedures. |
| Total Claim Cost | Sum of total claim cost across encounters. | Sum of TOTAL_CLAIM_COST. |
| Total Encounters | Number of encounter records. | Count of rows in encounters. |
| Zero Coverage % | Encounters with PAYER_COVERAGE equal to 0, divided by all encounters. | Zero Coverage Encounters / Total Encounters. |

## Data Quality Summary

- Source: six CSV files from AnalystLab Africa: 27,891 encounters, 974 patients, 47,701 procedures, 10 payers, 1 organization
- Period: 2 Jan 2011 to 5 Feb 2022
- No duplicate records and no orphan keys
- All start and stop times valid; none stops before it starts
- Reason fields blank in 70% of encounters and 77% of procedures; not used
- 89 encounters have a claim cost of zero; coverage never exceeds claim cost
- 48.7% of encounters have zero recorded payer coverage
- 47.4% of encounters have no procedure record
- 51% of procedure start times fall outside their encounter's time window, so procedures are placed by encounter date
- Four procedure codes appear under more than one description
- No rows removed and no values replaced

## Limitations

- One hospital only; 2022 is partial (data ends 5 Feb)
- February 2014 has 1,179 encounters against about 200 in a typical month; the data does not explain it
- Zero payer coverage may mean uninsured, unrecorded or unpaid
- Procedure base cost is not claim cost and should not be added to it
- The median encounter lasts 15 minutes; of the 75 over 24 hours, 13 last over 30 days, so treat "over 24 hours" with care
- Currency is not stated
- Records appear synthetic or anonymised
- No clinical outcomes, so no clinical claims

## Method & Handling

- Tools: Power Query (loading and cleaning), DAX (measures), Excel and SQL (validation)
- Timestamps are UTC; duration = STOP minus START, in hours
- Encounters link to patients, payers, organisations and dates; procedures link to encounters
- Personal columns were removed from patients before loading
- Key totals were checked independently and filters tested
- Prepared by Kukoyi Zainab Omotola; AI assistance (Claude) was used for planning, profiling and documentation